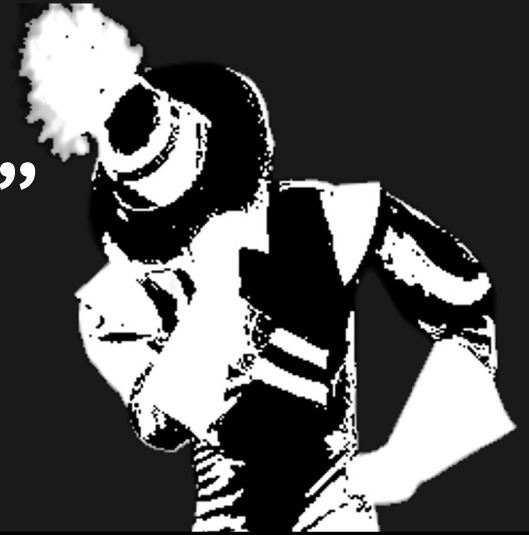


“Tell us your (marching arts) story”

A pilot LLM-supported phenomenology



Cory D. Meals

Elaine K. Waier

University of Houston | Moores School of Music

26th Clifford K. Madsen International Symposium

March 19, 2026 | Tallahassee, Florida

UNIVERSITY of
HOUSTON

KATHRINE G. McGOVERN
COLLEGE OF THE ARTS
Moores School of Music

Central Question

How do lived experience and social structure co-constitute each other within music ensembles?

- Ensemble participation is a culturally and socially meaningful lived experience (Abril, 2012; Adderley et al., 2003; Morrison, 2001)
- Social networks are central to ensemble function and experience (Matthews, 2017; Weren, 2016)
- Cognition and social networks co-constitute each other (Kleinbaum et al., 2020; Dunbar, 1998)
- Music ensemble social networks provide a window, and phenomenology allows us to explore the lived experiences within and between individuals, groups, and traditions

Pilot

Meals & Waier (in preparation)

RQ: What is it like to be a member of the marching arts?

Design

Hermeneutic phenomenology
(van Manen, 1997) with LLM
assistance
Retrospective audio reflections
Researcher positionality

Participants

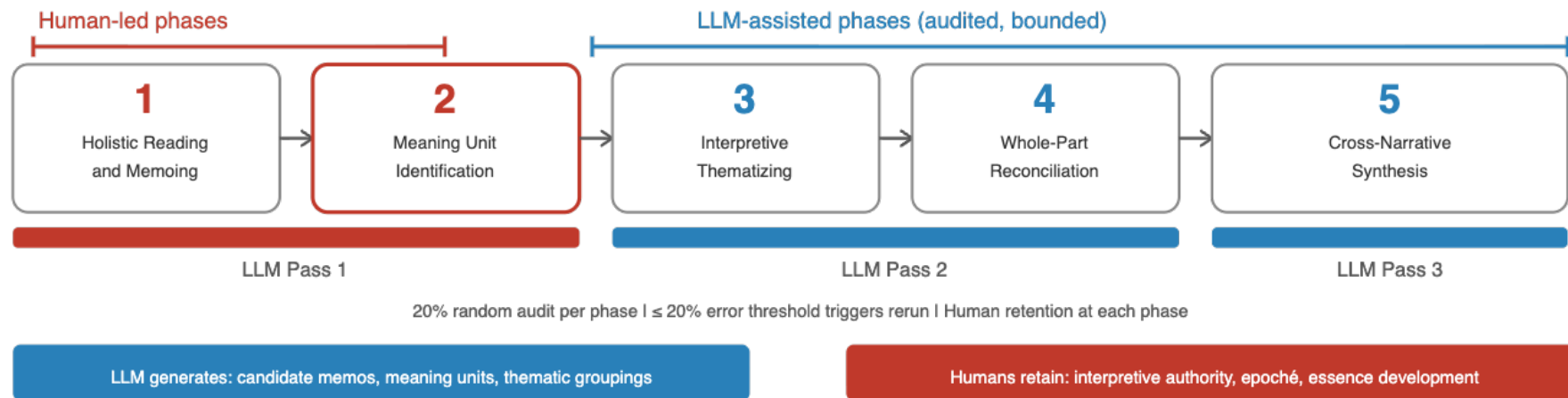
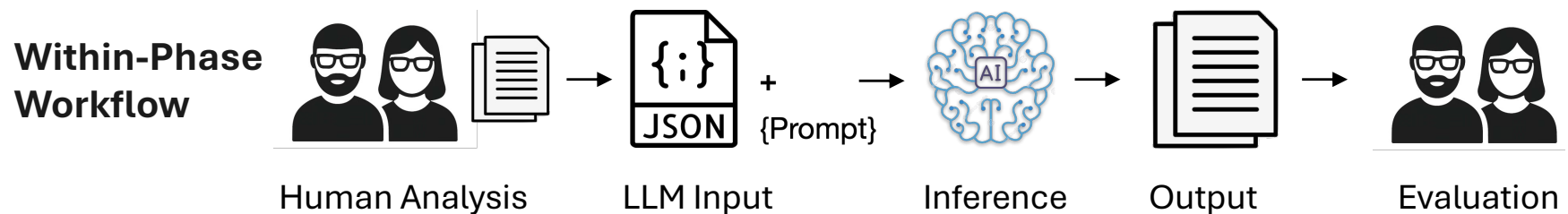
Sample: N = 6
HS band (6), College (4), DCI (4)*
Gender Identity: Male (4),
Female (2)
Experience Range: 1991–2025

Method

Hermeneutic
phenomenological analysis
(van Manen, 1997) *combined
with* seed-based, LLM-assisted
analysis
Human interpretive integration
and synthesis



Hybrid (Human-Machine) Workflow



Trustworthiness

Credibility

- Repeated engagement with transcripts across all five phases
- Team discussion and disagreement logs throughout analysis
- Competing readings of shared hardship resolved by returning to transcripts

Transferability

- Participants drawn from multiple settings and time periods (1991–2025)
- Interpretation kept close to participants' own language

Dependability

- Documented audit trail: analysis memos, decision logs, evidence tables

LLM Confirmability Procedure

- Two seed transcripts were fully analyzed by the research team
- LLM output treated as auditable analytic support only

20% Audit Threshold

- Random 20% of transcript-level outputs audited against the source at each LLM phase
- Phase rerun if >20% of audited outputs contained substantive error, omission, or hallucination
- Phase 2 required 2 iterations; Phase 4 required 2 iterations before stability criteria were met

Why This Matters

- LLM coding errors may be non-random with regard to participant characteristics (Ashwin et al., 2025)

Findings



Structure, Safety, and Chosen Commitment

Early membership: structured, supervised, protective. Adult oversight and school-based belonging.

Shifts toward chosen commitment, either college or corps settings, where all chose their involvement.



Shared Hardship, Intimacy, and Exposure

Difficulty as a connecting force: exhaustion, repetition, and public accountability lived together.

The same closeness that fosters intimacy can narrow into harm when intensity becomes coercive.



Growth Through Immersion in a Demanding Collective

Skill, awareness, and discipline emerged because the ensemble required them.

Growth understood relationally, not individually: “It was the system, and my operation within it, that helped me grow.”



Carrying the Ensemble Forward

Membership persists beyond and across seasons, but *what* endures is conditioned on the lived experience within the ensemble.

Conditional Essence

*Being a marching arts member is to exist within a highly organized, embodied, and social world in which belonging is achieved through shared effort, mutual exposure, and repeated alignment with others. **Membership is experienced as both sustaining and demanding.***

"That comfort, that shared experience... connected me so closely with these people that I didn't know were my brothers yet, but they were my brothers.."

— Participant 2

"We had enemies. We had people to blame. 'You're holding us back. You're the reason why we're getting 11th place.'"

— Participant 1

"Marching band led me to who I am today... in literally every sense of my life."

— Participant 5



Discussion

This pilot study...

- ... confirms and extends prior accounts of ensemble participation as deeply social and affective (Abril, 2012; Adderley et al., 2003; Morrison, 2001)
- ... specifies a temporal arc that prior work suggested: structured belonging → chosen commitment → identity that outlasts a given season (Matthews, 2017)
- ... connects to Cumberledge and colleagues (2024) on student leadership: both studies locate meaning in how participation shapes working with others, not in performance alone
- ... brings belonging and harm into the same frame: the same intensity that fosters intimacy can become coercive, consistent with Silveira & Hudson (2015) on hazing
- ... demonstrates that the marching arts may function as an embodied social world, not a discrete co-curricular activity



Limitations

Sample & Data

Sample ($N = 6$): Small human-coded reference set ($n = 2$), lack of quality examples for LLM

Sampling Frame: Sample concentrated in Texas with some Midwest/Northeast representation

Depth: Data are brief retrospective audio accounts instead of extended phenomenological interviews, limiting depth and textural richness

Cultural Differences: No HBCU representation, overrepresentation of drum corps, which weakens the conditional essence

Method & Interpretation

- LLM-assisted phases remain vulnerable to hallucination, over-patterning, and smoothing of marginal meanings
- Because participants' experiences consistently overlap emergent traditions, distinctions remain underdeveloped
- Positionality: research team experience rooted in corps-style traditions; performer perspective in college marching band missing

Scope of the Claim

The contribution is not that AI can conduct phenomenology. It is that, under tightly bounded conditions, machine assistance may help extend analysis without displacing human judgment.

Next Steps

Marching Arts Focus

Study 2: Comparing Analytic Pathways (in progress)

- Pilot transcripts, three workflows: human-only, human-machine, machine-only
- Three analysis layers: embedding-based semantic similarity, fidelity index, qualitative interpretation of findings
- Central question: What are workflow output differences within levels of semantic similarity, phenomenological fidelity, and overall quality?

Study 3: Scaled Deployment (planned)

- Larger sample, distributed snowball sampling, focus on including HBCU and non-corps-style traditions
- Refined hybrid methodology from Study 2; fine-tuned model grounded in high-quality, human-coded data

Future Research

- Validation Study, using Study 3 participants and the fine-tuned model
- Multi-group, longitudinal, social network embedded phenomenology



Methodological Contribution

Under tightly bounded conditions, LLM assistance may extend phenomenological analysis across many participants without displacing human interpretive authority.

Operational definition of “bounded”

- Human researchers served as the interpretive foundation at every phase
- LLM outputs were never accepted as final analytic determinations
- Seed cases provided a human-coded reference point for auditing model output
- Adversarial prompting and iterative stability testing before any output was retained
- Audit trail documented every decision: memos, disagreement logs, evidence tables

“Tell us your (marching arts) story”

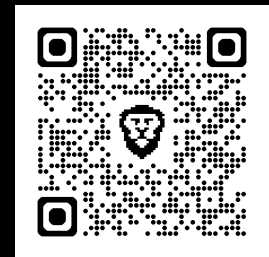
A pilot LLM-supported phenomenology



Marching arts membership is experienced as an embodied, demanding, and deeply social world. It frames and forms how people learn to live and grow with others under pressure. Its impact persists long after a season ends.

This finding emerged from a bounded human-machine workflow in which an LLM extended analytic reach without replacing human interpretive judgment.

Cory D. Meals | cdmeals@central.uh.edu
Elaine Waier | ewaier@cougarnet.uh.edu
University of Houston | Moores School of Music



Slide deck and
additional resources

UNIVERSITY of
HOUSTON

KATHRINE G. MCGOVERN
COLLEGE OF THE ARTS
Moores School of Music